

Xilong Zhou

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Education

- Ph.D., Computer Science, Texas A&M University.** Aug 2018 – Aug 2024
Advisor: Dr. Ergun Akleman and Dr. Nima Kalantari,
Dissertation: Towards Practical and Robust Material Acquisition and Generation.
- M.S.[†], Visualization Department, Texas A&M University. Aug 2016 – May 2018
Advisor: Dr. Ergun Akleman,
[†] Transfer to Computer Science Ph.D. Program.
- M.S., Petroleum Engineering, Texas A&M University.** Aug 2014 – Aug 2016
Advisor: Dr. Jenn-Tai Liang,
Thesis: Bulk and Adsorption Properties of Surfactant Entrapped in Polyelectrolyte Complex Nanoparticles.
- B.E., Petroleum Engineering, China University of Petroleum (Beijing).** Sep 2010 – Jun 2014

Research Interests

World Models, Generative Graphics, Video Generation, Neural Rendering, Novel View Synthesis, Inverse Rendering, and Relighting

Experiences

- Postdoc Researcher, Max Planck Institute for Informatics.** Sep 2024 – Present
Advisor: Dr. Christian Theobalt.
Topics: Generative AI, Video Editing, Neural Rendering, and Relighting.
- Research Intern, Adobe Research.** Feb – Jul 2024
Advisor: Dr. Miloš Hašan.
Topics: Realistic Diffusion-based Material Generation.
- Research Intern, Meta Reality Lab.** Aug 2022 – Apr 2023
Advisor: Dr. Chakravarty R. Alla Chaitanya.
Topics: Lightweight Novel View Synthesis.
- Research Intern, Adobe Research.** May – Aug 2022
Advisor: Dr. Miloš Hašan.
Topics: Realistic GAN-based Material Generation.
- Research Intern, Adobe Research.** May – Aug 2021
Advisor: Dr. Miloš Hašan.
Topics: Semi-procedural Material Priors & Editable Material Generation.

Publications

* Equal Contribution.

Preprints

- [1] **Xilong Zhou**, Bao-Huy Nguyen, Zheng Zeng, Jacob Munkberg, Jon Hasselgren, Thomas Leimkühler, Nima Kalantari, Miloš Hašan, Christian Theobalt. AlbedoEdit: Unified Instance-Level Video Editing with Albedo Guidance. Arxiv 2026.

Journal/Conference Articles

- [2] Pramod Rao, Jérémy Riviere, **Xilong Zhou**, Abhijeet Ghosh, Abhimitra Meka, Thabo Beeler, Marc Habermann, Christian Theobalt, Delio Vicini. Differentiable Polarized Path Tracing. *ECCV 2026*.
- [3] Timo Teufel, **Xilong Zhou**, Umar Iqbal, Jan Kautz, Marc Habermann, Vladislav Golyanik, Christian

Theobalt. D-Rex: Diffusion Rendering for Relightable Expressive Avatars. *ECCV 2026*.

[4] **Xilong Zhou**, Jianchun Chen*, Pramod Rao*, Timo Teufel, Linjie Lyu, Tigran Minasian, Oleksandr Sotnychenko, Xiao-Xiao Long, Marc Habermann, and Christian Theobalt. OLATverse: A Large-scale Real-world Object Dataset with Precise Lighting Control. *CVPR 2026 (Oral, Best Paper Award Candidate)*.

[5] **Xilong Zhou***, Bao-Huy Nguyen*, Loïc Magne, Vladislav Golyanik, Thomas Leimkühler, and Christian Theobalt. Splat the Net: Radiance Fields with Splattable Neural Primitives. *International Conference on Learning Representations (ICLR) 2026*.

[6] **Xilong Zhou**, Pedro Figueiredo, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Yiwei Hu, and Nima Khademi Kalantari. RealMat: Realistic Materials with Diffusion and Reinforcement Learning. *Computer Graphics Forum 2026*.

[7] Pramod Rao, **Xilong Zhou***, Abhimitra Meka*, Gereon Fox, Mallikarjun B R, Fangneng Zhan, Tim Weyrich, Bernd Bickel, Hanspeter Pfister, Wojciech Matusik, Thabo Beeler, Mohamed Elgharib, Marc Habermann, and Christian Theobalt. 3DPR: Single Image 3D Portrait Relighting with Generative Priors. *SIGGRAPH Asia 2025*.

[8] Avinash Paliwal, **Xilong Zhou**, Andrii Tsarov, and Nima Kalantari. PanoDreamer: Optimization-Based Single Image to 360 3D Scene With Diffusion. *SIGGRAPH Asia 2025*.

[9] Timo Teufel*, Pulkit Gera*, **Xilong Zhou**, Umar Iqbal, Pramod Rao, Jan Kautz, Vladislav Golyanik, and Christian Theobalt. HumanOLAT: A Large-Scale Dataset for Full-Body Human Relighting and Novel-View Synthesis. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) 2025*.

[10] Avinash Paliwal, **Xilong Zhou**, Wei Ye, Jinhui Xiong, Rakesh Ranjan, and Nima Kalantari. RI3D: Few-Shot Gaussian Splatting With Repair and Inpainting Diffusion Priors. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) 2025*.

[11] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Yannick Hold-Geoffroy, Kalyan Sunkavalli, and Nima Khademi Kalantari. PhotoMat: A Material Generator Learned from Single Flash Photos. *SIGGRAPH 2023*.

[12] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Kalyan Sunkavalli, and Nima Khademi Kalantari. A Semi-Procedural Convolutional Material Prior. *Computer Graphics Forum (Eurographics 2023)*.

[13] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Kalyan Sunkavalli, and Nima Khademi Kalantari. TileGen: Tileable, Controllable Material Generation and Capture. *SIGGRAPH Asia 2022*.

[14] **Xilong Zhou** and Nima Khademi Kalantari. Look-Ahead Training with Learned Reflectance Loss for Single-Image SVBRDF Estimation. *ACM Transactions on Graphics (SIGGRAPH Asia 2022)*.

[15] **Xilong Zhou** and Nima Khademi Kalantari. Adversarial Single-Image SVBRDF Estimation with Hybrid Training. *Computer Graphics Forum (Eurographics 2021)*.

[16] **Xilong Zhou**, Jenn-Tai Liang, Corbin D Andersen, Jiajia Cai, and Ying-Ying Lin. Enhanced Adsorption of Anionic Surfactants on Negatively Charged Quartz Sand Grains Treated with Cationic Polyelectrolyte Complex Nanoparticles. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 553, 397-405, 2018.

Academic Service

Activity: Primary Organizer of “See the World in a Different Light” (CVPR Workshop 2026).

Reviewer: SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, ICCV, TOG, IEEE TVCG, IEEE TPAMI, CGF, Computers & Graphics.

Selected Awards

Travel Grant, CSCE Department, Texas A&M University.	2023
National First Prize of National Petroleum Engineering Design Competition.	2013
Honorable Mention of International Mathematical Contest in Modeling.	2013
National Second Prize of National Mathematics Modeling Contest.	2012

Media Coverage

SIGGRAPH Thesis Fast Forward , Ph.D. Dissertation.	<i>2024</i>
Two Minutes Paper , Paper [11].	<i>2023</i>

Teaching

Co-Instructor , Graphics Seminar, Saarland University.	<i>Summer 2025</i>
Teaching Assistant , Machine Learning, Texas A&M University.	<i>Fall 2020, Spring 2022</i>
Teaching Assistant , Analysis of Algorithms, Texas A&M University.	<i>Summer 2019, Fall 2021</i>
Teaching Assistant , Programming, Texas A&M University.	<i>Spring 2021</i>
Teaching Assistant , Data Structure and Algorithm, Texas A&M University.	<i>Spring 2019</i>
Teaching Assistant , Discrete Structure for Computing, Texas A&M University.	<i>Fall 2019, Fall 2018</i>
Teaching Assistant , Computer for Visualization, Texas A&M University.	<i>Spring/Summer 2017</i>
Teaching Assistant , Formation Evaluation, Texas A&M University.	<i>Spring 2016</i>
Teaching Assistant , Unconventional Oil and Gas, Texas A&M University.	<i>Fall 2015</i>